Getting to Know Your Data

Attribute types, statistical description, visualization and proximity measures

Asif Zaman
Founder, Corpusian

Source: J. Han, M. Kamber & J. Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann, 2011 — Chapter 2

Roadmap

1. Why study the data first?
2. Data objects and attribute types
3. Basic statistical descriptions
4. Data visualization
5. Measuring similarity and dissimilarity
6. Summary and check-your-understanding

Learning outcomes

- Classify attributes and state which operations are meaningful
- Compute and interpret centre, spread and outlier fences
- Read boxplots, histograms, quantile, Q-Q and scatter plots
- Name the four families of multidimensional visualization
- Choose and compute a proximity measure for any data type

Why Study the Data First?

Section 1

Reconnaissance before mining

Every algorithm carries assumptions: *k*-means assumes averages are meaningful; *k*-NN assumes the distance reflects real resemblance. Violate them and the algorithm still answers — quietly and wrongly.

Four questions to ask first

1. What **kinds** of attributes do we have?
2. What do the **values** look like — centre, spread, skew, outliers?
3. Can we **see** the structure?
4. How **similar** are objects to one another?

Feeds into preprocessing Outliers → cleaning
Scale differences → normalization
Redundant attributes → reduction

Data Objects and Attribute Types

Section 2

Terminology

- A **data object** represents an entity — customer, patient, transaction.
  Also: sample, example, instance, data point, tuple, record.
- An **attribute** is a data field describing a characteristic of an object.
  Also: dimension (warehousing), feature (ML), variable (statistics).
- The attributes of an object form its **attribute (feature) vector**.
- One attribute $\Rightarrow$ *univariate*; two $\Rightarrow$ *bivariate*; many $\Rightarrow$ *multivariate*.

Key idea The **type** of an attribute is defined by the relationships among its values — and it decides which statistics, plots and distances are legitimate.

Nominal and binary attributes

Nominal (categorical)

- Symbols or names, **no order**: blood group, occupation, department
- Numeric codes are just labels
- Only meaningful centre: **mode**

> **Key idea** Student ID and postal code are *nominal*, although written in digits.

Binary nominal with two states, 0/1

- **Symmetric**: both outcomes equally important
  (e.g. gender)
- **Asymmetric**: one outcome matters more
  (e.g. positive test for a rare disease, purchase of an item)
- Convention: code the rarer, more important outcome as 1

Ordinal and numeric attributes

Ordinal

- Meaningful **order**, unknown magnitude between values
- Drink size, academic rank, Likert rating, letter grade
- Centre: **median** or mode

Numeric (quantitative)

- **Interval-scaled**: equal units, *no true zero*
  °C, calendar year — differences only
- **Ratio-scaled**: inherent zero
  kelvin, weight, age, income — ratios meaningful

20 °C is not "twice as hot" as 10 °C; Tk 80,000 is twice Tk 40,000.

Discrete vs continuous (ML view): finite or countably infinite values (number of children) vs real values (height). Binary is a special case of discrete.

Attribute types at a glance

Type	Order	Operations	Centre	Example
Nominal	no	$=, \neq$	mode	blood group, occupation
Binary (sym.)	no	$=, \neq$	mode	gender
Binary (asym.)	no	$=, \neq$ (1 matters)	mode	disease test result
Ordinal	yes	$=, \neq, <, >$	median, mode	grade, satisfaction
Interval	yes	$+, -$	mean, median, mode	°C, year
Ratio	yes	$+, -, \times, \div$	mean, median, mode	weight, income, age

Basic Statistical Descriptions

Section 3

Central tendency: the mean

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad \bar{x}_w = \frac{\sum_{i=1}^{N} w_i x_i}{\sum_{i=1}^{N} w_i}$$

- **Algebraic**: built from sum() and count() — cheap to maintain incrementally
- Sensitive to extreme values
- **Trimmed mean**: drop a small percentage (e.g. 2%) at each end before averaging

Measure classes **Distributive**: sum, count, min, max
Algebraic: mean, variance (from distributive)
Holistic: median, mode (need all data)

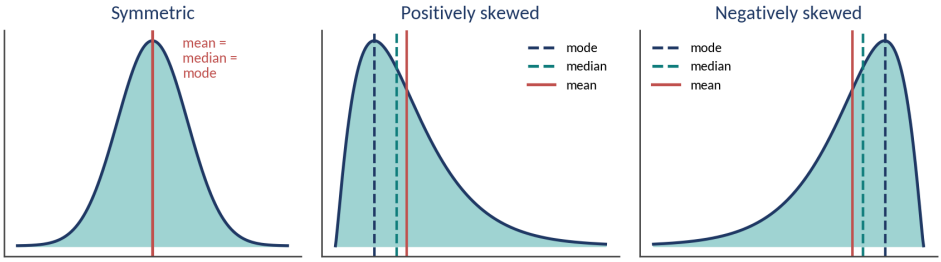
Median, mode and midrange

- **Median**: middle of the ordered data (average the two middle values when N is even). Robust to outliers; right choice for skewed and ordinal data. *Holistic*.
- **Grouped data** — interpolate within the median interval:

$$\text{median} \approx L_1 + \left(\frac{N/2 - (\sum \text{freq})_l}{\text{freq}_{\text{median}}} \right) \times \text{width}$$

- **Mode**: most frequent value; data can be uni-, bi-, tri- or multimodal. Empirically, for moderately skewed data: $\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$.
- **Midrange**: $(\min + \max)/2$ — cheap, but hostage to the extremes.

Symmetry and skew



mean = median = mode

mode $<$ median $<$ mean
long right tail (income)

mean $<$ median $<$ mode
long left tail

Worked Example 1 — central tendency

Exam scores of 12 students (sorted)

42, 48, 55, 58, 61, 61, 64, 67, 70, 72, 75, 98

- Mean $= 771/12 =$ **64.25**
- Median $= (61 + 64)/2 =$ **62.5**
- Mode = **61** (unimodal)
- Midrange $= (42 + 98)/2 =$ **70.0**
- Trimmed mean (drop 1 each end) $= 631/10 =$ **63.1**

Interpretation mode (61) $<$ median (62.5) $<$ mean (64.25)
$\Rightarrow$ mildly **positively skewed**, driven by the score of 98. Trimming pulls the mean toward the median.

Worked Example 2 — median from grouped data

Age group	Frequency	Cumulative
0–20	120	120
20–40	300	420
40–60	260	680
60–80	120	800

- $N/2 = 400 \Rightarrow$ median interval is **20–40**
- $L_1 = 20,\ (\sum \text{freq})_l = 120,\ \text{freq}_{\text{median}} = 300,\ \text{width} = 20$

$$\text{median} \approx 20 + \frac{400 - 120}{300} \times 20 = \mathbf{38.67}\ \text{years}$$

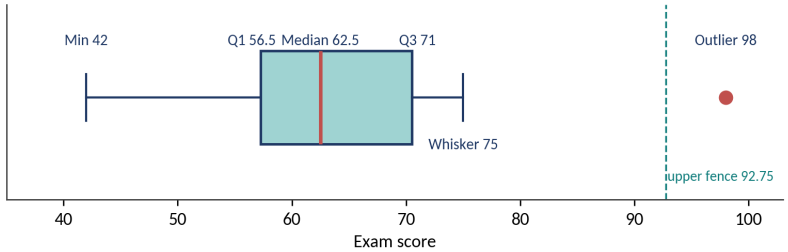
Dispersion: quantiles, IQR and the five-number summary

- **Range** $= \max - \min$
- **Quantiles** split ordered data into equal-size groups: 2-quantile = median; 4-quantiles = **quartiles** Q_1, Q_2, Q_3; 100-quantiles = percentiles
- **Interquartile range**: $IQR = Q_3 - Q_1$ (spread of the middle half)
- **Five-number summary**: Min, Q_1, Median, Q_3, Max

Suspected outlier if outside $\left[Q_1 - 1.5 \times IQR,\quad Q_3 + 1.5 \times IQR\right]$

Convention Quartile definitions differ between packages. We use **median-of-halves**: Q_1 = median of the lower half, Q_3 = median of the upper half. Always state your convention.

Worked Example 3 — boxplot of the exam scores



- Lower half $\Rightarrow Q_1 = (55 + 58)/2 = \mathbf{56.5}$
- Upper half $\Rightarrow Q_3 = (70 + 72)/2 = \mathbf{71.0}$
- $IQR = \mathbf{14.5}$; fences $\mathbf{34.75}$, $\mathbf{92.75}$
- $98 > 92.75 \Rightarrow$ suspected outlier
- Whisker stops at 75
- Summary: 42, 56.5, 62.5, 71, 98

Variance and standard deviation

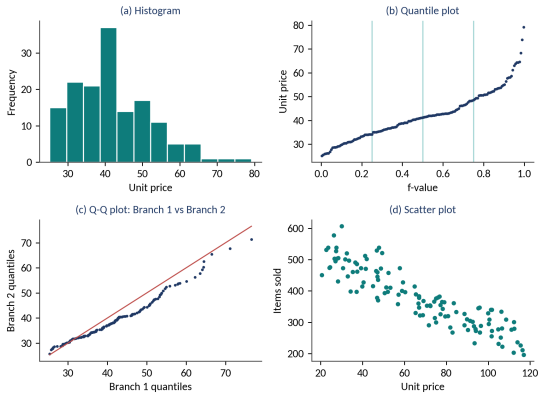
$$\sigma^2 \;=\; \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2 \;=\; \left(\frac{1}{N}\sum_{i=1}^{N}x_i^2\right) - \bar{x}^2, \qquad \sigma \;=\; \sqrt{\sigma^2}$$

- Small σ: values cluster near the mean; $\sigma = 0$ only if all values are equal
- Chebyshev: at least $1 - 1/k^2$ of values lie within $k\sigma$ of the mean — $\geq 75\%$ within 2σ, whatever the shape
- Second form shows variance is **algebraic** $\Rightarrow$ scalable
- Sample estimate divides by $N - 1$; the text (and these slides) use N

Example 3 (cont.)

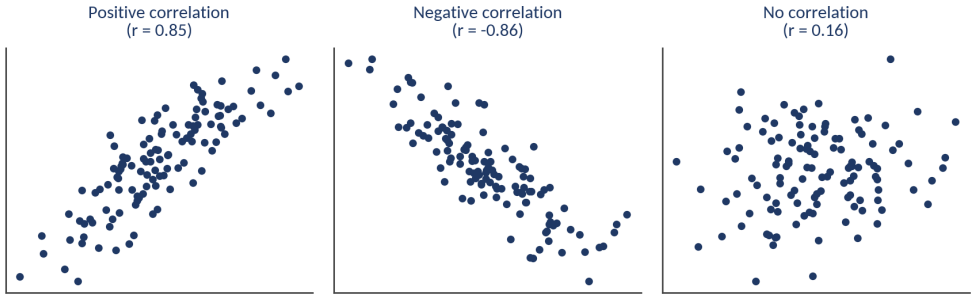
Exam scores: $\sigma^2 = \mathbf{188.35}$, $\sigma = \mathbf{13.72}$ (with $N - 1$: $s = 14.33$)

Graphic displays of basic statistics



- **Histogram**: counts per bucket; bar chart for nominal data. Shape depends on bin width.
- **Quantile plot**: sorted x_i against $f_i = \dfrac{i - 0.5}{N}$; every value shown.
- **Q-Q plot**: quantiles of one distribution vs another; agreement $\Leftrightarrow$ 45° line.
- **Scatter plot**: two numeric attributes; reveals correlation, clusters, outliers.

Reading correlation from scatter plots



Key idea Points rising left-to-right ⇒ positive correlation; falling ⇒ negative; shapeless cloud ⇒ no linear relationship. Formal tests (Pearson's r, χ^2) come with data integration in Chapter 3.

Data Visualization

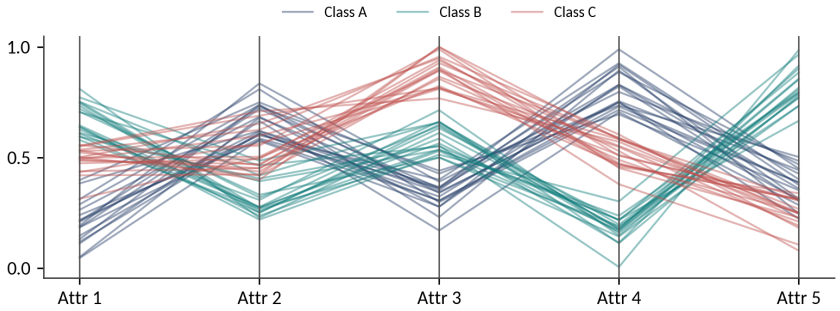
Section 4

Four families of multidimensional visualization

Family	Core idea	Examples / caveats
Pixel-oriented	One coloured pixel per value, one window per attribute	Space-filling curves, circle segments; shows very many objects
Geometric projection	Project onto 2-D/3-D layouts	Scatter-plot matrix (n^2 panels), parallel coordinates (clutter)
Icon-based	Encode attributes as icon features	Chernoff faces ($\approx$18 variables; feature salience biases reading), stick figures
Hierarchical	Nest subsets of dimensions	Worlds-within-worlds, tree-maps, dimensional stacking

Complex data & relations: tag clouds for text; node–link graphs for social or citation networks.

Parallel coordinates



- *n* equally spaced vertical axes, one per attribute; each object is a polyline
- Scales to many dimensions — but many objects produce clutter
- Here the three classes separate cleanly on Attributes 1, 2 and 4

Similarity and Dissimilarity

Section 5

Proximity and the two data structures

- **Similarity**: larger when objects are alike; **dissimilarity**: 0 when identical. Normalized to $[0, 1]$: $\text{sim}(i, j) = 1 - d(i, j)$

Data matrix (two-mode)

$$\begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{bmatrix}_{n \times p}$$

objects $\times$ attributes

Dissimilarity matrix (one-mode)

$$\begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \cdots & 0 \end{bmatrix}_{n \times n}$$

objects $\times$ objects; symmetric, store lower triangle

Nominal attributes

$$d(i,j) = \frac{p-m}{p} \qquad p = \text{number of attributes, } m = \text{number of matches}$$

Worked Example 4 — (Programme, Phone OS, Division)

O1 = (CSE, Android, Dhaka), O2 = (EEE, iOS, Dhaka), O3 = (CSE, Android, Khulna), O4 = (ICT, Android, Dhaka)

$$\begin{bmatrix} 0 & & & \\ 0.67 & 0 & & \\ 0.33 & 1.00 & 0 & \\ 0.33 & 0.67 & 0.67 & 0 \end{bmatrix}$$

O2 and O3 share nothing ($d = 1$); O1 is equally close to O3 and O4.

Binary attributes

	$j=1$	$j=0$	sum
$i=1$	q	r	$q+r$
$i=0$	s	t	$s+t$
sum	$q+s$	$r+t$	p

Symmetric: $d(i,j) = \dfrac{r+s}{q+r+s+t}$

Asymmetric: $d(i,j) = \dfrac{r+s}{q+r+s}$

Jaccard: $\mathrm{sim}(i,j) = \dfrac{q}{q+r+s}$

Key idea For asymmetric attributes the negative matches t are dropped: two customers who both did *not* buy caviar are not thereby alike.

Worked Example 5 — asymmetric binary

Attributes: fever, cough, test-1, test-2, test-3, test-4 (1 = present/positive)

Rahim = (1,0,1,0,0,0) Karim = (1,0,1,0,1,0) Salma = (1,1,0,0,0,0)

Pair	q	r	s	d (asym.)	Jaccard
Rahim–Karim	2	0	1	**0.33**	0.67
Rahim–Salma	1	1	1	**0.67**	0.33
Karim–Salma	1	2	1	**0.75**	0.25

Key idea Treated (wrongly) as symmetric, Rahim–Karim would give $1/6 = 0.17$ — the three shared negatives inflate their similarity.

Numeric data: normalize first

Attributes in large units (income in taka) swamp those in small units (age in years).

Min-max: $v' = \dfrac{v - \min_A}{\max_A - \min_A}(\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$

z-score: $v' = \dfrac{v - \bar{A}}{\sigma_A}$

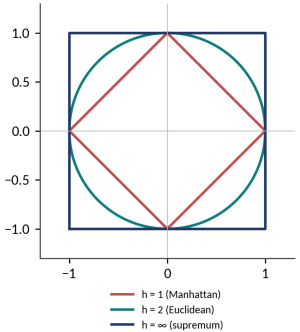
Example Exam scores: min-max of 70 onto $[0,1] = (70-42)/(98-42) = 0.5$; z-score of 98 $= (98 - 64.25)/13.72 = 2.46$.

The Minkowski family

$$d(i,j) \;=\; \Big(\sum_{f=1}^{p} |x_{if} - x_{jf}|^{h}\Big)^{1/h}, \quad h \geq 1$$

- $h = 1$: **Manhattan** (city block) $\sum_f |x_{if} - x_{jf}|$
- $h = 2$: **Euclidean** $\sqrt{\sum_f (x_{if} - x_{jf})^2}$
- $h \to \infty$: **supremum** ($L_{\max}$, Chebyshev) $\max_f |x_{if} - x_{jf}|$

Metric properties: non-negativity; $d(i,i) = 0$; symmetry; triangle inequality $d(i,j) \leq d(i,k) + d(k,j)$.



Unit "circles" for $h = 1, 2, \infty$

Worked Example 6 — Minkowski distances

$x_1 = (2, 7, 3)$, $x_2 = (5, 3, 3) \quad \Rightarrow \quad |x_1 - x_2| = (3, 4, 0)$

Manhattan ($h = 1$)	$3 + 4 + 0 = \mathbf{7}$
Euclidean ($h = 2$)	$\sqrt{9 + 16 + 0} = \sqrt{25} = \mathbf{5}$
Minkowski ($h = 3$)	$(27 + 64 + 0)^{1/3} = 91^{1/3} \approx \mathbf{4.50}$
Supremum ($h \to \infty$)	$\max(3, 4, 0) = \mathbf{4}$

Key idea $L_1 \geq L_2 \geq L_3 \geq L_\infty$: raising h gives progressively more weight to the largest single difference.

Ordinal attributes

1. Replace each value x_{if} by its rank $r_{if} \in \{1, \ldots, M_f\}$
2. Map onto $[0, 1]$ so every attribute carries equal weight:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$ then apply any numeric distance

Worked Example 7 — poor $<$ fair $<$ good $<$ excellent ($M = 4$)

Normalized ranks: 0, 0.33, 0.67, 1.
O1 = excellent (1), O2 = fair (0.33), O3 = good (0.67), O4 = excellent (1)
$d(O1, O2) = 0.67,\ d(O1, O3) = 0.33,\ d(O1, O4) = 0,\ d(O2, O3) = 0.33,\ d(O2, O4) = 0.67,\ d(O3, O4) = 0.33$

Mixed attribute types

$$d(i,j) \quad = \quad \frac{\sum_{f=1}^{p} \delta_{ij}^{(f)}\, d_{ij}^{(f)}}{\sum_{f=1}^{p} \delta_{ij}^{(f)}}$$

- $\delta_{ij}^{(f)} = 0$ if x_{if} or x_{jf} is missing, or both are 0 for an asymmetric binary f; otherwise 1
- **Numeric**: $d_{ij}^{(f)} = \dfrac{|x_{if} - x_{jf}|}{\max_h x_{hf} - \min_h x_{hf}}$
- **Nominal / binary**: 0 if equal, 1 otherwise
- **Ordinal**: compute z_{if}, then treat as numeric

Worked Example 8 — mixed types

Pair	Nominal	Ordinal	Numeric	$d(i, j)$
O1, O2	1	0.667	0.548	0.738
O1, O3	0	0.333	0.452	**0.262**
O1, O4	1	0	0.405	0.468
O2, O3	1	0.333	1.000	**0.778**
O2, O4	1	0.667	0.143	0.603
O3, O4	1	0.333	0.857	0.730

Key idea O1–O3 most similar (same programme, adjacent grades, close hours); O2–O3 least similar.

Cosine similarity and the Tanimoto coefficient

Documents and baskets are **long, sparse** vectors; shared zeros should not count as resemblance.

$$\text{sim}(\mathbf{x},\mathbf{y}) = \frac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\,\|\mathbf{y}\|} \qquad \text{Tanimoto}(\mathbf{x},\mathbf{y}) = \frac{\mathbf{x}\cdot\mathbf{y}}{\mathbf{x}\cdot\mathbf{x}+\mathbf{y}\cdot\mathbf{y}-\mathbf{x}\cdot\mathbf{y}}$$

Worked Example 9

$\mathbf{x}=(3,0,2,1,0,4)$, $\mathbf{y}=(1,0,3,0,2,2)$: $\mathbf{x}\cdot\mathbf{y}=17$, $\|\mathbf{x}\|=\sqrt{30}$, $\|\mathbf{y}\|=\sqrt{18}\Rightarrow$ sim $\approx \mathbf{0.73}$
Binary $(1,0,1,1,0,1)$ vs $(1,0,1,0,1,1)$: Tanimoto $=3/(4+4-3)=\mathbf{0.60}$ (= Jaccard)

Cosine is a similarity, *not* a metric: $1-\cos$ can violate the triangle inequality.

Choosing a proximity measure

Data type	Measure	Remark
Nominal	$(p - m)/p$	or expand to asymmetric binary
Symmetric binary	$(r + s)/(q + r + s + t)$	both outcomes equally informative
Asymmetric binary	$(r + s)/(q + r + s)$; Jaccard	ignore negative matches
Numeric	Manhattan, Euclidean, Minkowski, supremum	normalize first
Ordinal	rank $\to (r - 1)/(M - 1) \to$ numeric	equal weight across attributes
Mixed	δ-weighted average	handles missing values
Sparse vectors	cosine; Tanimoto (binary)	ignores shared zeros

Summary

Section 6

Take-home messages

- **Classify before you compute**: attribute type governs legitimate statistics, plots and distances.
- **Centre**: mean, median, mode, midrange — their ordering reveals skew; median and trimmed mean resist outliers.
- **Spread**: range, quartiles, IQR, five-number summary, σ; the $1.5 \times IQR$ rule is a quick outlier screen.
- **Displays**: histograms, quantile and Q-Q plots, scatter plots expose shape and relationships.
- **Visualization**: pixel-oriented, geometric projection, icon-based, hierarchical.
- **Proximity**: match the measure to the data — and normalize numeric attributes first.

Check your understanding

1. Is a Likert rating of 1–5 interval-scaled? Defend your answer.
2. For 8, 12, 15, 15, 18, 21, 24, 27, 30, 55: give the five-number summary and flag any outliers.
3. Compute the Manhattan, Euclidean and supremum distances between $(1, 4, 6, 2)$ and $(3, 1, 6, 7)$.
4. Why is the Jaccard coefficient, not simple matching, appropriate for market-basket data?
5. Why can Euclidean distance mislead when comparing a long and a short document on the same topic?

Answers: (2) 8, 15, 19.5, 27, 55; fences -3, $45 \Rightarrow 55$ is an outlier. (3) 10, $\sqrt{38} \approx 6.16$, 5.

Further reading

- J. Han, M. Kamber and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed., Morgan Kaufmann, 2011 — Ch. 2 and Ch. 3.
- J. W. Tukey, *Exploratory Data Analysis*, Addison-Wesley, 1977.
- E. R. Tufte, *The Visual Display of Quantitative Information*, 2nd ed., Graphics Press, 2001.

Thank you

Questions and discussion

Asif Zaman · Founder, Corpusian